

# Validity Analysis: MATHDIAL

Target context: Indian Metropolitan Professional Teachers (Grades 1–8)

Overall risk: **HIGH**

## Deployment Context

**Use case:** Deployment scenario: A GPT-4 model is implemented in a one-on-one tutoring session, where its responses are augmented with a human tutor’s input. The tutor evaluates the pedagogical quality of the combined response, particularly in terms of how well it provides guidance. The goal is to assess whether the augmented GPT-4 responses are acceptable to the tutor.

Domain: Educational tutoring

Setting: Mobile application / enterprise software

**Target population:** High-end professional teacher teaching grade 1-8 in major metropolitan cities in India, such as Delhi and Mumbai, who is fluent in English.

## Dimension Scores

| Dimension         | Score | Rating           | Confidence |
|-------------------|-------|------------------|------------|
| Input Ontology ✓  | 4     | Minor gaps       | high       |
| Input Content △   | 1     | Serious concern  | high       |
| Input Form ✓      | 5     | Strong alignment | high       |
| Output Ontology △ | 2     | Significant gaps | high       |
| Output Content △  | 1     | Serious concern  | high       |
| Output Form ✓     | 5     | Strong alignment | high       |
| Average           | 3.0   |                  |            |

△ = highest concern    ✓ = strongest dimension

## Dimension Key

| Abbr. | Dimension       | Definition                                                                                                   |
|-------|-----------------|--------------------------------------------------------------------------------------------------------------|
| IO    | Input Ontology  | Whether the benchmark’s test case categories cover the query types expected in deployment.                   |
| IC    | Input Content   | Whether individual datapoint content is culturally and contextually appropriate for the target region.       |
| IF    | Input Form      | Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.             |
| OO    | Output Ontology | Whether the benchmark’s output categories and scoring criteria reflect regionally valid decision boundaries. |
| OC    | Output Content  | Whether ground-truth labels align with the judgments of regional stakeholders.                               |
| OF    | Output Form     | Whether the expected output modality matches regional deployment needs and accessibility.                    |

## Overall Summary

MathDial presents a structurally sound but culturally narrow evaluation framework for AI tutoring quality. For the Indian Metropolitan Professional Teachers (Grades 1–8) deployment, three dimensions are well-aligned (Input Form, Output Form, and — given user confirmation that the eight categories suffice — Input Ontology), while three dimensions raise serious

concerns. Input Content (HIGH user priority) shows fundamental misalignment: the LLM-simulated student exhibits verbose, peer-register, reasoning-out-loud behavior diametrically opposed to the deferential brief low-verbalization style the user identifies as characteristic of Indian metro classrooms, math word problems are entirely US-origin, and LLM looping artifacts produce unnatural student behavior. Output Content (HIGH user priority) is the deepest concern: ground-truth labels were produced by predominantly UK/US annotators explicitly instructed toward Socratic non-directive scaffolding, with  $\kappa=0.34$  inter-annotator agreement on Equitable Tutoring even within the Western pool — and the user explicitly expects Indian professional teachers' judgments to systematically diverge on direct-correction vs. probing and exam-readiness weighting. Output Ontology (MODERATE) embeds a constructivist anti-telling structural orientation (Telling@k as negative metric, Telling as last-resort) that conflicts with Indian professional teachers' view of direct correction as a primary pedagogical tool, and the user's prioritization of 'providing guidance' is not commensurately weighted in the three co-equal evaluation dimensions. The English text-only modality match is the benchmark's strongest validity asset for this deployment.

## Practical Guidance

### What This Benchmark Measures

MathDial can support evaluation of whether GPT-4-augmented tutoring responses adhere to a Western constructivist scaffolding framework on English-language multi-step math word problems. The benchmark's strongest contributions for the Indian metro deployment are its move-level taxonomy (IO, score 4) — providing a structured vocabulary of Focus, Probing, Telling, and Generic that the user has accepted as sufficient — and its form match (IF, OF both score 5), which means modality and language are not a barrier. It can measure surface-level dialogue coherence and mathematical correctness reliably.

### Construct Depth

The benchmark probes pedagogical quality shallowly for the Indian metro context. Input Content (score 1) means the student behavior the benchmark exposes evaluators to is systematically unrepresentative of Indian classrooms, so any judgments are anchored to dialogues that would not occur with real Indian students. Output Content (score 1) means the comparator ground-truth quality standard encodes Western Socratic norms the user explicitly identifies as misaligned. Output Ontology (score 2) means even when the right categories are present, their relative weighting and the framing of Telling-as-failure conflict with Indian professional pedagogical priorities. Together, IC and OC failures mean the benchmark cannot deeply probe the construct of 'good guidance for Indian metro professional teachers' — it probes 'good guidance for Western-trained constructivist tutors,' which is a different construct.

### What Else You Need

(a) Re-annotation of a representative subsample (e.g., 100–200 dialogues) by Indian metro professional teachers to produce parallel ground-truth labels, enabling measurement of  $\kappa$  between Western and Indian labels and identifying systematic divergence dimensions (addresses OC). (b) Augmentation or replacement of student simulation with dialogues reflecting Indian classroom interaction norms — brief, deferential, low-verbalization students — to address IC; this could draw on transcripts of Indian classrooms or use stakeholder-elicited student personas. (c) Reweighting or replacement of the Equitable Tutoring dimension with a 'Providing Guidance' dimension operationalized to credit appropriate directive instruction, addressing OO. (d) Inversion of Telling@k framing for the Indian context, treating early-but-appropriate Telling as a positive rather than negative outcome where it matches teacher-rated quality. (e) Cross-validation with PARIKSHA-style multilingual evaluation findings [WEB-18] to calibrate expectations of LLM-evaluator alignment with Indian human judgments.

| ID     | Evidence                                                                                                                                                                                     |
|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| WEB-18 | PARIKSHA finding: LLM evaluators align less with human evaluations on culturally nuanced responses ( <a href="https://arxiv.org/html/2406.15053v1">https://arxiv.org/html/2406.15053v1</a> ) |

# Input Ontology (IO)

Score: 4/5 — Minor gaps | Confidence: high

Benchmark: MATHDIAL | Context: Indian Metropolitan Professional Teachers (Grades 1–8)

**Input Ontology definition:** Whether the benchmark's test case categories cover the query types expected in deployment.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

## Justification

The user explicitly confirmed that curriculum-agnostic math errors and the benchmark's eight pedagogical categories are necessary and sufficient for the deployment task [A1, A3], substantially reducing ontological misalignment risk. The taxonomy is theoretically grounded in Reiser's (2004) scaffolding framework distinguishing structure (Focus) from problematize (Probing) moves [Q36, Q37], with Telling and Generic as additional categories [Q40, Q41]. The dataset confirms transparent application of the four-move taxonomy across all sampled dialogues, providing structured pedagogical vocabulary [DATASET-D2, DATASET-D3]. Confusion type coverage spans six algebra-misconception categories [DATASET-D7, DATASET-D8, DATASET-D10]. Minor concerns remain: the taxonomy's structural privileging of scaffolding over directive correction [Q154, Q155] embeds a constructivist orientation that, while not omitting categories the user requires, frames the relationships among them in a way that may not match Indian metro teaching priorities. The benchmark also explicitly omits meta-cognitive support and rapport-building [Q112], though the user did not flag these as required.

| ID   | Evidence                                                                                                                                                                                                                                                                                                                                                                       |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q36  | "We base the first two on the work of Reiser (2004), who suggest that scaffolding strategies can be split into two main categories: structure and problematize." (p.5)                                                                                                                                                                                                         |
| Q37  | "These form the basis for the Focus and Probing moves employed in our study." (p.5)                                                                                                                                                                                                                                                                                            |
| Q40  | "This is called Telling." (p.5)                                                                                                                                                                                                                                                                                                                                                |
| Q41  | "Finally, turns that just serve as conversational elements and have limited pedagogical value are classed as Generic." (p.5)                                                                                                                                                                                                                                                   |
| Q112 | "Inspired by previous work in scaffolding, we acknowledge our focus is on a subset of common teaching moves. However, this does not cover all the goals of human tutors, such as meta-cognitive support or building rapport with a student." (p.10)                                                                                                                            |
| Q154 | "Most importantly, scaffolding questions that are productive for long-term learning are Focus and Probing." (p.17)                                                                                                                                                                                                                                                             |
| Q155 | "On the other hand, Telling represents giving out the partial or full answer to the student and should be mostly used when a student is stuck." (p.17)                                                                                                                                                                                                                         |
| D2   | Hi Rishi. Can you tell me how many hours it would take Apple to run 24 miles?" ... "Good. How many hours would it take Mac?" ... "How much longer does it take Apple?" ... "How much longer is that in minutes?" ... "So how much faster is Mac then Apple in minutes?" — Six consecutive probing questions guiding student to answer; entirely Socratic, no direct correction |
| D3   | Carmela has $600 \times 2 + 50 = 1250$ ... "It is 1250. Not 2450" ... $600 + 950 + 1250 = 2800$ — Teacher resorts to Telling only after multiple Focus moves fail; Telling framed as reluctant fallback                                                                                                                                                                        |
| D7   | Student: Sure. In the first half of the year, Nicki ran for 26 weeks. Since I wanted to calculate how many miles she ran per week, I divided the total number of weeks by 2 to get the average number of miles she ran per week. — Simulated student produces unprompted multi-clause explanation of own reasoning process                                                     |
| D8   | Student: I divided the total number of desserts (126) by the number of people in the family (7) to find out how many desserts each person would get (18). Then, I divided 18 by 3, since there are 3 types of desserts, to find out how many of each dessert each person would get (6). — Unprompted full explanation of two-step reasoning from student                       |
| D10  | Student profile: Winnie is a 7th grade student. She struggle most with understanding what the problem is asking them to do. — US-style student name and brief English-language confusion profile; no Indian cultural markers                                                                                                                                                   |

## Strengths

- Eight pedagogical categories user-confirmed as necessary and sufficient for the deployment task [A3]
- Theoretically anchored in scaffolding learning sciences (Reiser 2004) [Q36, Q37]

- Fine-grained move labels systematically applied to every teacher utterance, enabling multi-dimensional pedagogical evaluation [DATASET-D2, DATASET-D3]
- Broad coverage of student confusion types relevant to Grades 1–8 math reasoning [DATASET-D7, DATASET-D8, DATASET-D10]

| ID  | Evidence                                                                                                                                                                                                                                                                                                                                                                      |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q36 | "We base the first two on the work of Reiser (2004), who suggest that scaffolding strategies can be split into two main categories: structure and problematize." (p.5)                                                                                                                                                                                                        |
| Q37 | "These form the basis for the Focus and Probing moves employed in our study." (p.5)                                                                                                                                                                                                                                                                                           |
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| D8  | Student: I divided the total number of desserts (126) by the number of people in the family (7) to find out how many desserts each person would get (18). Then, I divided 18 by 3, since there are 3 types of desserts, to find out how many of each dessert each person would get (6). — Unprompted full explanation of two-step reasoning from student                      |
| D10 | Student profile: Winnie is a 7th grade student. She struggle most with understanding what the problem is asking them to do. — US-style student name and brief English-language confusion profile; no Indian cultural markers                                                                                                                                                  |

## Checklist

### Checklist item definitions:

| ID   | Assessment Question                                            |
|------|----------------------------------------------------------------|
| IO-1 | Identify test case categories required for regional deployment |
| IO-2 | Check if taxonomy omits regionally relevant categories         |
| IO-3 | Check if taxonomy includes irrelevant categories               |
| IO-4 | Document category gaps harming content validity                |

### Checklist responses:

**IO-1:** User confirmed curriculum-agnostic math errors and the eight pedagogical categories suffice for the Indian metro deployment [A1, A3]. Required test categories include Focus, Probing, Telling, Generic moves and conceptual/procedural error confusions, all present in the benchmark.

**IO-2:** No regionally-relevant categories are identified by the user as missing. The benchmark omits meta-cognitive support, rapport-building [Q112], and visual instructional practices [Q113], but the user did not flag these as required for the deployment.

**IO-3:** No categories are identified as irrelevant to the regional context. All four move types (Focus, Probing, Telling, Generic) appear in real classroom practice and Indian metro teachers can evaluate them, though the framing of Telling as a last resort [Q155] may be at odds with Indian pedagogical norms (handled under OO).

**IO-4:** No major content-validity gaps identified at the ontology level given user confirmation of sufficiency [A3]. Residual concern is that the taxonomy's hierarchical valuation (scaffolding productive, Telling last-resort) [Q154, Q155] is itself a value-laden ontological choice.

| ID   | Evidence                                                                                                                                                                                                                                            |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q42  | "Table 2 lists finer-grained intents for each of these four categories along with a set of accompanying examples." (p.5)                                                                                                                            |
| Q112 | "Inspired by previous work in scaffolding, we acknowledge our focus is on a subset of common teaching moves. However, this does not cover all the goals of human tutors, such as meta-cognitive support or building rapport with a student." (p.10) |

|      |                                                                                                                                                        |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q113 | "Moreover, text tutoring limits teachers' use of additional instructional practices such as drawings." (p.10)                                          |
| Q154 | "Most importantly, scaffolding questions that are productive for long-term learning are Focus and Probing." (p.17)                                     |
| Q155 | "On the other hand, Telling represents giving out the partial or full answer to the student and should be mostly used when a student is stuck." (p.17) |

## Information Gaps

- Whether the user's acceptance of the eight categories was made with full awareness of how Telling is structurally devalued in the framework

## Requires Expert Verification

- Whether Indian metro teachers, after reviewing benchmark dialogues, would still affirm the eight categories as sufficient or would flag missing categories (e.g., explicit demonstration, drill, exam-strategy guidance)

## Remediation

**Gap 1: Although user accepted the eight categories, the taxonomy embeds a hierarchical valuation (scaffolding 'productive', Telling 'last resort') that may not survive scrutiny once Indian teachers see the benchmark dialogues.**

**Recommendation:** Conduct a confirmatory taxonomy-elicitation session with 5–10 Indian metro teachers after they review 20+ benchmark dialogues, asking whether they would add categories (e.g., explicit demonstration, drill, exam-strategy guidance) or merge existing ones. Update taxonomy if needed before deployment.

# Input Content (IC)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: MATHDIAL | Context: Indian Metropolitan Professional Teachers (Grades 1–8)

**Input Content definition:** Whether individual datapoint content is culturally and contextually appropriate for the target region.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

## Justification

Input content is the highest-priority concern (user-flagged HIGH) and the dataset analysis confirms severe construct-irrelevant variance. The student simulation is calibrated to Western tutoring conventions [Q32, Q34] with documented limitations in representing human learning [Q106]. Empirically, the LLM-simulated students invariably produce verbose, multi-sentence, reasoning-out-loud responses [DATASET-D6, DATASET-D7, DATASET-D9] — the polar opposite of the brief, deferential, low-verbalization style the user identifies as characteristic of Indian metro classrooms [A2]. Students even use casual peer-register openers ('Hi, I'm doing well. How can I help you?') [DATASET-D24] inconsistent with deferential Indian student-teacher interactions. Math word problems are entirely US-origin from GSM8k [Q103, Q118] with US currency and scenarios (recycling deposits, temp agencies) [DATASET-D15, DATASET-D16, DATASET-D17] and no Indian cultural grounding. Nominal name diversity [Q123, DATASET-D11] does not extend to behavioral diversity. Additional concern: simulated student loops on identical wrong answers across multiple teacher corrections [DATASET-D13, DATASET-D14] — an LLM artifact not representative of any real student population. Authors explicitly acknowledge teachers may have adapted to model patterns rather than exhibiting naturalistic behavior [Q108]. The MathDial dialogues thus encode interaction dynamics fundamentally misaligned with Indian metro classrooms at the level of every individual datapoint.

| ID   | Evidence                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q32  | "We use InstructGPT (text-davinci-003) (Ouyang et al., 2022) to generate student turns." (p.5)                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Q34  | "The prompt contains the MWP, the initial student confusion, as well as the student profile which explains the type of confusion and persona of the student." (p.5)                                                                                                                                                                                                                                                                                                                                                       |
| Q103 | "Our dataset consists of ca. 3k tutoring conversations grounded in math word problems from GSM8k." (p.9)                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Q106 | "In this work, we used an LLM to simulate student confusion. However, we acknowledge that these models have a limited understanding of human learning and this is a key limitation in our dataset – certain kinds of student confusions may be under- or over-represented in our dataset." (p.9)                                                                                                                                                                                                                          |
| Q108 | "Furthermore, in our setup, teachers were interacting with an LLM role-playing as a student. However, it is possible that some teachers might have learned to interact with the student model in a different way than they would do in the classroom." (p.10)                                                                                                                                                                                                                                                             |
| Q118 | "While the problems in GSM8k are simple enough to be understood quickly by teachers, they remain challenging for students, who among others have to deal with equations or percentages." (p.14)                                                                                                                                                                                                                                                                                                                           |
| Q123 | "To build a dataset that would reflect students of various backgrounds, we use numerous student names associated with their given pronouns." (p.15)                                                                                                                                                                                                                                                                                                                                                                       |
| D6   | Jia: Sure. I started by figuring out Maxwell's age in 2 years... So, if Maxwell is twice her age in 2 years, he will be 4 years old. Then, I subtracted 2 from 4 to get the answer that Maxwell is currently 2 years old. — Simulated student gives detailed, multi-sentence explanation of reasoning — verbose, articulate, reasoning-out-loud style                                                                                                                                                                     |
| D7   | Student: Sure. In the first half of the year, Nicki ran for 26 weeks. Since I wanted to calculate how many miles she ran per week, I divided the total number of weeks by 2 to get the average number of miles she ran per week. — Simulated student produces unprompted multi-clause explanation of own reasoning process                                                                                                                                                                                                |
| D9   | Student: I started by adding the number of racers at the beginning of the race (50) and the number of racers who joined after 20 minutes (30), which gave me a total of 80 racers. Then, I multiplied that number by two... Finally, I subtracted the total number of racers who finished the race (130) from the total number of racers after 30 minutes (160) to get the total number of people who dropped before finishing the race, which was 80. — Three-step unprompted reasoning walkthrough by simulated student |
| D11  | Student profile: Rishi is a 7th grade student. — Student name "Rishi" is South Asian; however student behavior pattern (detailed equation explanation) remains Western-expressive                                                                                                                                                                                                                                                                                                                                         |

|            |                                                                                                                                                                                                                                                                                                                                                                |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>D13</b> | "Student: That's correct. Carmela has $\$1200 \times 2 + \$50 = \$2450$ ." (after teacher corrects with "Carmela has $600 \times 2 + 50 = 1250$ ") — Student continues to repeat prior error even after told correct answer; simulated LLM student fails to update believably                                                                                  |
| <b>D14</b> | "Student: Oh, I see. That means he gave Bobby 3 gumballs. He gave him 4 times as many as he gave Alisha, but then subtracted 5. So, $3 = 4 \times 2 - 5$ ." (repeated three times with nearly identical text after teacher corrections) — Simulated student persists in same wrong answer verbatim across multiple teacher moves; artificial LLM loop behavior |
| <b>D15</b> | question: Cecil has \$600. Catherine has \$250 less than twice as much as Cecil... — US-dollar amounts and US-style name inventory (Cecil, Catherine, Carmela); no Indian cultural grounding                                                                                                                                                                   |
| <b>D16</b> | question: Jack collects all his neighbors' recycling and sorts out the cans and bottles to return for the deposit. He gets 10 cents per can. — US-specific recycling-deposit context; cultural scenario without Indian analogue                                                                                                                                |
| <b>D17</b> | question: 20% of the job candidates at a temp agency have experience with Excel. 70% of the candidates are only willing to work day shifts. — US labor-market scenario (temp agency, day-shift context); no Indian cultural equivalent                                                                                                                         |
| <b>D24</b> | Teacher: (generic)Hi mariana, how are you?" / "Student: Hi, I'm doing well. How can I help you? — Casual social opener from student; register is informal and conversational in Western peer-style rather than deferential                                                                                                                                     |

## Strengths

- Mathematical content (multi-step word problems, percentages, rates, equations) is broadly familiar to Indian Grade 1–8 teachers per user confirmation [A2]
- Single nominally South Asian name appears in sample (Rishi) [DATASET-D11], indicating the pipeline is not exclusionary at the name level
- Authors transparently document student simulation limitations [Q106, Q108]

| ID          | Evidence                                                                                                                                                                                                                                                                                         |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Q106</b> | "In this work, we used an LLM to simulate student confusion. However, we acknowledge that these models have a limited understanding of human learning and this is a key limitation in our dataset – certain kinds of student confusions may be under- or over-represented in our dataset." (p.9) |
| <b>Q108</b> | "Furthermore, in our setup, teachers were interacting with an LLM role-playing as a student. However, it is possible that some teachers might have learned to interact with the student model in a different way than they would do in the classroom." (p.10)                                    |
| <b>D11</b>  | Student profile: Rishi is a 7th grade student. — Student name "Rishi" is South Asian; however student behavior pattern (detailed equation explanation) remains Western-expressive                                                                                                                |

## Checklist

### Checklist item definitions:

| ID          | Assessment Question                                                        |
|-------------|----------------------------------------------------------------------------|
| <b>IC-1</b> | Do inputs require region-specific cultural/geographic/dialectal knowledge? |
| <b>IC-2</b> | Does culturally sensitive content align with target culture?               |
| <b>IC-3</b> | Flag inputs assuming Western-specific knowledge                            |
| <b>IC-4</b> | Regional annotator recruitment for culturally sensitive instances          |
| <b>IC-5</b> | Document content issues harming content validity                           |

### Checklist responses:

**IC-1:** Yes — the deployment requires student dialogues that reflect Indian classroom interaction norms (deferential, brief, low-verbalization students) per user elicitation [A2]. The benchmark does not provide region-specific cultural or interactional content.

**IC-2:** No — culturally sensitive content (US monetary units, US labor scenarios, US-default names) does not align with Indian deployment context [DATASET-D15, DATASET-D16, DATASET-D17]. Nominal diversity does not extend to interaction style [DATASET-D11].

**IC-3:** Yes — inputs require US/Western cultural frames (recycling-deposit systems [DATASET-D16], temp-agency labor markets [DATASET-D17], dollar denominations [DATASET-D15]) that do not transfer naturally to Indian classroom contexts.



**IC-4:** Authors did not recruit Indian-specific annotators to validate cultural appropriateness; India was a minority contributor to the annotator pool [Q27] and not separately consulted for IC validation.

**IC-5:** Critical content-validity gaps documented: (a) simulated student verbosity diverges from Indian deferential norms [A2, DATASET-D6, DATASET-D7, DATASET-D9]; (b) no Indian cultural grounding in MWP's [DATASET-D15, DATASET-D16, DATASET-D17]; (c) LLM looping artifacts unrepresentative of any real student [DATASET-D13, DATASET-D14]; (d) procedure-heavy and language-mediated misunderstandings characteristic of Indian L2-English math instruction underrepresented [A2].

| ID     | Evidence                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q27    | "The majority of annotators are nationals of the UK, followed by the USA, Canada, Australia, India, and Germany, with a median age of 39 years." (p.4)                                                                                                                                                                                                                                                                                                                                                                    |
| Q32    | "We use InstructGPT (text-davinci-003) (Ouyang et al., 2022) to generate student turns." (p.5)                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Q106   | "In this work, we used an LLM to simulate student confusion. However, we acknowledge that these models have a limited understanding of human learning and this is a key limitation in our dataset – certain kinds of student confusions may be under- or over-represented in our dataset." (p.9)                                                                                                                                                                                                                          |
| Q108   | "Furthermore, in our setup, teachers were interacting with an LLM role-playing as a student. However, it is possible that some teachers might have learned to interact with the student model in a different way than they would do in the classroom." (p.10)                                                                                                                                                                                                                                                             |
| D6     | Jia: Sure. I started by figuring out Maxwell's age in 2 years... So, if Maxwell is twice her age in 2 years, he will be 4 years old. Then, I subtracted 2 from 4 to get the answer that Maxwell is currently 2 years old. — Simulated student gives detailed, multi-sentence explanation of reasoning — verbose, articulate, reasoning-out-loud style                                                                                                                                                                     |
| D7     | Student: Sure. In the first half of the year, Nicki ran for 26 weeks. Since I wanted to calculate how many miles she ran per week, I divided the total number of weeks by 2 to get the average number of miles she ran per week. — Simulated student produces unprompted multi-clause explanation of own reasoning process                                                                                                                                                                                                |
| D9     | Student: I started by adding the number of racers at the beginning of the race (50) and the number of racers who joined after 20 minutes (30), which gave me a total of 80 racers. Then, I multiplied that number by two... Finally, I subtracted the total number of racers who finished the race (130) from the total number of racers after 30 minutes (160) to get the total number of people who dropped before finishing the race, which was 80. — Three-step unprompted reasoning walkthrough by simulated student |
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| D14    | "Student: Oh, I see. That means he gave Bobby 3 gumballs. He gave him 4 times as many as he gave Alisha, but then subtracted 5. So, $3 = 4 \times 2 - 5$ ." (repeated three times with nearly identical text after teacher corrections) — Simulated student persists in same wrong answer verbatim across multiple teacher moves; artificial LLM loop behavior                                                                                                                                                            |
| D15    | question: Cecil has \$600. Catherine has \$250 less than twice as much as Cecil... — US-dollar amounts and US-style name inventory (Cecil, Catherine, Carmela); no Indian cultural grounding                                                                                                                                                                                                                                                                                                                              |
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| D17    | question: 20% of the job candidates at a temp agency have experience with Excel. 70% of the candidates are only willing to work day shifts. — US labor-market scenario (temp agency, day-shift context); no Indian cultural equivalent                                                                                                                                                                                                                                                                                    |
| WEB-15 | Systematic review: majority Indic LLM datasets are translations and models fare poorly on socio-cultural tasks ( <a href="https://arxiv.org/html/2501.13912v1">https://arxiv.org/html/2501.13912v1</a> )                                                                                                                                                                                                                                                                                                                  |
| WEB-18 | PARIKSHA finding: LLM evaluators align less with human evaluations on culturally nuanced responses ( <a href="https://arxiv.org/html/2406.15053v1">https://arxiv.org/html/2406.15053v1</a> )                                                                                                                                                                                                                                                                                                                              |

## Information Gaps

- Quantitative magnitude of acceptable mismatch in interaction dynamics — at what threshold do simulated dialogues become unusable for Indian teacher evaluation?
- Whether Indian teachers would refuse to evaluate the dialogues at all or simply rate them differently

## Requires Expert Verification

- Whether L2-English-mediated mathematical misunderstandings would change quality judgments substantially



- Whether dialogues featuring nominal Indian names but Western-expressive behavior produce uncanny-valley effects for Indian evaluators
- 

## Remediation

**Gap 1: Simulated student verbosity, Western-expressive register, US-origin word problems, and LLM looping artifacts mean every datapoint embeds construct-irrelevant variance for Indian metro evaluation.**

**Recommendation:** Generate or curate a supplementary subset of dialogues featuring deferential, brief, low-verbalization Indian student personas — either by re-prompting an LLM with Indian-classroom-calibrated personas or by sourcing real Indian classroom transcripts. Include Indian currency (rupees) and culturally familiar scenarios in math problems. Validate that the new dialogues are rated 'typical of Indian sixth grader' by Indian teacher annotators.

**Gap 2: LLM-simulated students exhibit non-naturalistic looping behavior (repeating identical wrong answers verbatim across multiple teacher corrections), an artifact of temperature sampling not representative of any real student population.**

**Recommendation:** Filter out dialogues exhibiting student-loop pathology (e.g., where consecutive student turns share >80% verbatim overlap after a teacher correction) before using MathDial for Indian evaluation. This reduces the construct-irrelevant variance contribution from LLM artifacts independent of the cultural mismatch issue.

# Input Form (IF)

Score: 5/5 — Strong alignment | Confidence: high

Benchmark: MATHDIAL | Context: Indian Metropolitan Professional Teachers (Grades 1–8)

**Input Form definition:** Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

## Justification

MathDial is English text-only [Q1, Q32], and the deployment context is confirmed English text-only on mobile/enterprise platforms with no code-switching, script variation, or vernacular content. There is no modality, script, or language-encoding mismatch. Dataset analysis confirms all 18 sampled examples are pure English prose with no Hindi, Hinglish, or transliteration [DATASET-D6, DATASET-D15]. A safety filter using Perspective API removes toxic content [Q157, Q158]. Minor surface-level imperfections in teacher utterances (spelling/typing errors) [DATASET-D19] are noted but do not affect form validity. The deployment is Indian metro with English-medium instruction, and metro infrastructure supports the input modality [WEB-3, WEB-4, WEB-8].

| ID    | Evidence                                                                                                                                                                                                                                                                                                                                              |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q1    | "We describe how we use this framework to collect MATHDIAL, a dataset of 3k one-to-one teacher-student tutoring dialogues grounded in multi-step math reasoning problems." (p.1)                                                                                                                                                                      |
| Q32   | "We use InstructGPT (text-davinci-003) (Ouyang et al., 2022) to generate student turns." (p.5)                                                                                                                                                                                                                                                        |
| Q157  | "As we are interested in real educational use cases for our tutoring system, we apply a safety filter to filter out conversations with any sensitive content." (p.18)                                                                                                                                                                                 |
| Q158  | "In particular, we use the Perspective API to filter out conversations containing toxic content (<1%)." (p.18)                                                                                                                                                                                                                                        |
| D6    | Jia: Sure. I started by figuring out Maxwell's age in 2 years... So, if Maxwell is twice her age in 2 years, he will be 4 years old. Then, I subtracted 2 from 4 to get the answer that Maxwell is currently 2 years old. — Simulated student gives detailed, multi-sentence explanation of reasoning — verbose, articulate, reasoning-out-loud style |
| D15   | question: Cecil has \$600. Catherine has \$250 less than twice as much as Cecil... — US-dollar amounts and US-style name inventory (Cecil, Catherine, Carmela); no Indian cultural grounding                                                                                                                                                          |
| D19   | Teacher: Luca do you think yooou could check your answer again please? maye try and think..." / "Teacher: was the total three trimes the amount oof his contribution before oor after he contributed? — Multiple spelling and typographic errors in teacher utterances; affects naturalness of benchmark dialogue                                     |
| WEB-3 | Delhi has 3,000+ English-medium schools ( <a href="https://ezyschooling.com/english-medium-schools-in-delhi">https://ezyschooling.com/english-medium-schools-in-delhi</a> )                                                                                                                                                                           |
| WEB-4 | Delhi government Schools of Excellence operate in English medium ( <a href="https://www.schoolmykids.com/schools/list-of-schools-of-excellence-in-delhi-india">https://www.schoolmykids.com/schools/list-of-schools-of-excellence-in-delhi-india</a> )                                                                                                |
| WEB-8 | National school internet connectivity 63.5% per UDISE+ 2024-25 ( <a href="https://educationforallindia.com/analysis-of-udise-2024-25-data-by-prof-arun-c-mehta/">https://educationforallindia.com/analysis-of-udise-2024-25-data-by-prof-arun-c-mehta/</a> )                                                                                          |

## Strengths

- Perfect form alignment: both benchmark and deployment are English text-only [Q1, Q32]
- All sampled dialogues confirmed monolingual English with no script issues [DATASET-D6, DATASET-D15]
- Safety filtering applied to remove toxic content [Q157, Q158]
- Metro infrastructure supports text-based interfaces [WEB-3, WEB-8]

| ID  | Evidence                                                                                                                                                                         |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q1  | "We describe how we use this framework to collect MATHDIAL, a dataset of 3k one-to-one teacher-student tutoring dialogues grounded in multi-step math reasoning problems." (p.1) |
| Q32 | "We use InstructGPT (text-davinci-003) (Ouyang et al., 2022) to generate student turns." (p.5)                                                                                   |

|              |                                                                                                                                                                                                                                                                                                                                                       |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Q157</b>  | "As we are interested in real educational use cases for our tutoring system, we apply a safety filter to filter out conversations with any sensitive content." (p.18)                                                                                                                                                                                 |
| <b>Q158</b>  | "In particular, we use the Perspective API to filter out conversations containing toxic content (<1%)." (p.18)                                                                                                                                                                                                                                        |
| <b>D6</b>    | Jia: Sure. I started by figuring out Maxwell's age in 2 years... So, if Maxwell is twice her age in 2 years, he will be 4 years old. Then, I subtracted 2 from 4 to get the answer that Maxwell is currently 2 years old. — Simulated student gives detailed, multi-sentence explanation of reasoning — verbose, articulate, reasoning-out-loud style |
| <b>D15</b>   | question: Cecil has \$600. Catherine has \$250 less than twice as much as Cecil... — US-dollar amounts and US-style name inventory (Cecil, Catherine, Carmela); no Indian cultural grounding                                                                                                                                                          |
| <b>WEB-3</b> | Delhi has 3,000+ English-medium schools ( <a href="https://ezyschooling.com/english-medium-schools-in-delhi">https://ezyschooling.com/english-medium-schools-in-delhi</a> )                                                                                                                                                                           |
| <b>WEB-8</b> | National school internet connectivity 63.5% per UDISE+ 2024-25 ( <a href="https://educationforallindia.com/analysis-of-udise-2024-25-data-by-prof-arun-c-mehta/">https://educationforallindia.com/analysis-of-udise-2024-25-data-by-prof-arun-c-mehta/</a> )                                                                                          |

## Checklist

### Checklist item definitions:

| ID          | Assessment Question                                          |
|-------------|--------------------------------------------------------------|
| <b>IF-1</b> | Compare signal distributions between source and target       |
| <b>IF-2</b> | Check if regional infrastructure supports data capture specs |
| <b>IF-3</b> | Identify domain-specific form differences                    |
| <b>IF-4</b> | Document form mismatches harming external validity           |

### Checklist responses:

**IF-1:** No signal-distribution mismatch. Both source and target use English Latin-script natural language text [Q1, Q32, DATASET-D15].

**IF-2:** Yes — Indian metro infrastructure supports text-based AI tutoring interfaces; Delhi has 3,000+ English-medium schools [WEB-3] and Government Schools of Excellence operate in English medium [WEB-4]. National school internet connectivity is 63.5% [WEB-8] with metros expected higher.

**IF-3:** No domain-specific form differences. The benchmark and deployment both operate in text-based open-ended dialogue.

**IF-4:** No form mismatches identified. Minor surface noise in teacher utterances (typos) [DATASET-D19] is naturalistic and does not harm external validity for the deployment.

| ID           | Evidence                                                                                                                                                                                                                                                                                                         |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Q1</b>    | "We describe how we use this framework to collect MATHDIAL, a dataset of 3k one-to-one teacher-student tutoring dialogues grounded in multi-step math reasoning problems." (p.1)                                                                                                                                 |
| <b>Q32</b>   | "We use InstructGPT (text-davinci-003) (Ouyang et al., 2022) to generate student turns." (p.5)                                                                                                                                                                                                                   |
| <b>D15</b>   | question: Cecil has \$600. Catherine has \$250 less than twice as much as Cecil... — US-dollar amounts and US-style name inventory (Cecil, Catherine, Carmela); no Indian cultural grounding                                                                                                                     |
| <b>D19</b>   | Teacher: Luca do you think youu could check your answer again please? maye try and think..." / "Teacher: was the total three trimes the amount oof his contribution before oor after he contributed? — Multiple spelling and typographic errors in teacher utterances; affects naturalness of benchmark dialogue |
| <b>WEB-3</b> | Delhi has 3,000+ English-medium schools ( <a href="https://ezyschooling.com/english-medium-schools-in-delhi">https://ezyschooling.com/english-medium-schools-in-delhi</a> )                                                                                                                                      |
| <b>WEB-4</b> | Delhi government Schools of Excellence operate in English medium ( <a href="https://www.schoolmykids.com/schools/list-of-schools-of-excellence-in-delhi-india">https://www.schoolmykids.com/schools/list-of-schools-of-excellence-in-delhi-india</a> )                                                           |
| <b>WEB-8</b> | National school internet connectivity 63.5% per UDISE+ 2024-25 ( <a href="https://educationforallindia.com/analysis-of-udise-2024-25-data-by-prof-arun-c-mehta/">https://educationforallindia.com/analysis-of-udise-2024-25-data-by-prof-arun-c-mehta/</a> )                                                     |

# Output Ontology (OO)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: MATHDIAL | Context: Indian Metropolitan Professional Teachers (Grades 1–8)

**Output Ontology definition:** Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

## Justification

User-flagged MODERATE risk. The user accepted the eight categories as necessary and sufficient [A3], reducing one source of ontological gap. However, the benchmark's output taxonomy operationalizes Equitable Tutoring [Q97, Q98, Q166] with an explicitly anti-telling, constructivist orientation, treating Telling as a last resort [Q155] and using Telling@k as a negative metric [Q96] — directly conflicting with the Indian metro pedagogical norm of direct correction as a primary, not last-resort, strategy. The user identified 'providing guidance' as the single most critical dimension [A3], but the benchmark distributes evaluation across three co-equal dimensions (Coherence, Correctness, Equitable Tutoring) [Q97, Q166], not weighting guidance commensurately. Dataset analysis confirms structural embedding of these values: Telling is positioned as late-stage fallback after multiple Focus/Probing failures [DATASET-D3, DATASET-D5, DATASET-D27], and self-correctness annotations frame Telling as 'had to reveal' [DATASET-D5, DATASET-D23] — a quality-deficit framing rather than legitimate pedagogical strategy. Additionally, the benchmark's classification of teacher openings that fully restate correct setups as 'Generic' rather than 'Telling' [DATASET-D12, DATASET-D28] reveals taxonomy ambiguity Indian annotators might resolve differently. This is structural validity violation: the construct's structure (what counts as good guidance) is misrepresented for the target context.

| ID   | Evidence                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q96  | "Telling@k measures the percentage of conversations where the teacher explicitly tells the final answer before the student has reached it on their own within the first k turns." (p.7)                                                                                                                                                                                                                                                                                                                                         |
| Q97  | "Finally, we conduct a human evaluation according to three criteria: 1) Coherence: how coherent the teacher's response is with respect to the preceding dialogue, 2) Correctness: whether it is in itself correct, and 3) Equitable tutoring." (p.8)                                                                                                                                                                                                                                                                            |
| Q98  | "Equitable tutoring describes how well the model provides the student with room for exploring the problem and solution space." (p.8)                                                                                                                                                                                                                                                                                                                                                                                            |
| Q155 | "On the other hand, Telling represents giving out the partial or full answer to the student and should be mostly used when a student is stuck." (p.17)                                                                                                                                                                                                                                                                                                                                                                          |
| Q166 | "The following dimensions were rated by annotators: Coherence - "The response naturally follows up on the previous utterance and context and has no logical conflicts with the context." Correctness - "The response is factually and mathematically correct and respects the learning concepts being taught." Equitable tutoring - "The response gives a learning opportunity for the student by providing space for reflection, explanation, pointing to follow-up challenge, or engaging the student in other ways."" (p.18) |
| D3   | Carmela has $600 \times 2 + 50 = 1250$ ... "It is 1250. Not 2450" ... $600 + 950 + 1250 = 2800$ — Teacher resorts to Telling only after multiple Focus moves fail; Telling framed as reluctant fallback                                                                                                                                                                                                                                                                                                                         |
| D5   | No $32 - 7 = 25$ . So the difference is 25" ... "No. His total running time was 25 seconds. You need to subtract the amount of time it took him to run up the first half of the hill to find your answer — Telling moves are labeled and visible in transcript; student resolves confusion only after explicit revelation                                                                                                                                                                                                       |
| D12  | Teacher: Hi Riya. Shawn painted 9 red pebbles, 13 blue pebbles, and the 18 remaining pebbles into 3 equal groups of 6 yellow, 6 purple, and 6 green pebbles. Correct? — Teacher opens by recapping the correct setup — essentially pre-solving before asking student; minimal scaffolding effort                                                                                                                                                                                                                                |
| D23  | Teacher: (telling)You need to find out how many times 48 goes into 192" / "Teacher: (telling)192 divide by 48 =" / "Teacher: (telling)4 is the correct answer you don't need to calculate further — Three consecutive Telling moves before student accepts correct answer; demonstrates reluctant resort to direct instruction                                                                                                                                                                                                  |
| D27  | Teacher: (telling)19 - 14 = 5 minutes. — Single arithmetic Telling move that resolves the conversation; teacher had already spent multiple Focus moves before telling                                                                                                                                                                                                                                                                                                                                                           |

|     |                                                                                                                                                                                                                                                                                                                 |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| D28 | Teacher: (generic)Hi Riya. Shawn painted 9 red pebbles, 13 blue pebbles, and the 18 remaining pebbles into 3 equal groups of 6 yellow, 6 purple, and 6 green pebbles. Correct? — Opening move presents the full correct setup to student before posing the actual question; near-complete Telling at the outset |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## Strengths

- Eight pedagogical categories user-confirmed as necessary and sufficient [A3]
- Explicit operational definitions provided for each output dimension [Q166]
- Three-dimensional human evaluation framework offers some richness [Q97]
- Self-correctness annotations enable filtering by tutoring strategy used [DATASET-D5, DATASET-D23]

| ID   | Evidence                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q97  | "Finally, we conduct a human evaluation according to three criteria: 1) Coherence: how coherent the teacher's response is with respect to the preceding dialogue, 2) Correctness: whether it is in itself correct, and 3) Equitable tutoring." (p.8)                                                                                                                                                                                                                                                                            |
| Q166 | "The following dimensions were rated by annotators: Coherence - "The response naturally follows up on the previous utterance and context and has no logical conflicts with the context." Correctness - "The response is factually and mathematically correct and respects the learning concepts being taught." Equitable tutoring - "The response gives a learning opportunity for the student by providing space for reflection, explanation, pointing to follow-up challenge, or engaging the student in other ways."" (p.18) |
| D5   | No $32-7 = 25$ . So the difference is 25" ... "No. His total running time was 25 seconds. You need to subtract the amount of time it took him to run up the first half of the hill to find your answer — Telling moves are labeled and visible in transcript; student resolves confusion only after explicit revelation                                                                                                                                                                                                         |
| D23  | Teacher: (telling)You need to find out how many times 48 goes into 192" / "Teacher: (telling)192 divide by 48 =" / "Teacher: (telling)4 is the correct answer you don't need to calculate further — Three consecutive Telling moves before student accepts correct answer; demonstrates reluctant resort to direct instruction                                                                                                                                                                                                  |

## Checklist

### Checklist item definitions:

| ID   | Assessment Question                                          |
|------|--------------------------------------------------------------|
| OO-1 | Review output label categories for regional relevance        |
| OO-2 | Identify missing regionally specific categories              |
| OO-3 | Flag categories encoding non-regional values                 |
| OO-4 | Consider stakeholder-driven taxonomy redesign                |
| OO-5 | Document taxonomy issues harming structural/content validity |

### Checklist responses:

**OO-1:** Output label categories are user-accepted as relevant [A3], but the Equitable Tutoring dimension [Q98, Q166] encodes a constructivist anti-telling orientation that does not match Indian metro pedagogical priorities [A4].

**OO-2:** No new categories required per user [A3], but the deployment requires that 'providing guidance' be weighted as primary [A3], whereas the benchmark treats it as one of three co-equal dimensions [Q97].

**OO-3:** Yes — Equitable Tutoring [Q98, Q166] explicitly rewards withholding answers and giving room for exploration, encoding constructivist Western values misaligned with Indian direct-correction norms. Telling@k [Q96] frames direct instruction as a failure mode.

**OO-4:** Stakeholder-driven taxonomy reweighting (not redesign) is warranted: 'providing guidance' should be elevated as primary and Telling reframed as legitimate strategy rather than last-resort failure [A3, A4, DATASET-D5, DATASET-D23].

**OO-5:** Structural validity issue documented: the benchmark's scoring function encodes a Socratic anti-telling preference [Q154, Q155, Q96] that the user explicitly identifies as misaligned with Indian professional teacher pedagogy [A4].

| ID | Evidence |
|----|----------|
|----|----------|

|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q96  | "Telling@k measures the percentage of conversations where the teacher explicitly tells the final answer before the student has reached it on their own within the first k turns." (p.7)                                                                                                                                                                                                                                                                                                                                         |
| Q97  | "Finally, we conduct a human evaluation according to three criteria: 1) Coherence: how coherent the teacher's response is with respect to the preceding dialogue, 2) Correctness: whether it is in itself correct, and 3) Equitable tutoring." (p.8)                                                                                                                                                                                                                                                                            |
| Q98  | "Equitable tutoring describes how well the model provides the student with room for exploring the problem and solution space." (p.8)                                                                                                                                                                                                                                                                                                                                                                                            |
| Q154 | "Most importantly, scaffolding questions that are productive for long-term learning are Focus and Probing." (p.17)                                                                                                                                                                                                                                                                                                                                                                                                              |
| Q155 | "On the other hand, Telling represents giving out the partial or full answer to the student and should be mostly used when a student is stuck." (p.17)                                                                                                                                                                                                                                                                                                                                                                          |
| Q166 | "The following dimensions were rated by annotators: Coherence - "The response naturally follows up on the previous utterance and context and has no logical conflicts with the context." Correctness - "The response is factually and mathematically correct and respects the learning concepts being taught." Equitable tutoring - "The response gives a learning opportunity for the student by providing space for reflection, explanation, pointing to follow-up challenge, or engaging the student in other ways."" (p.18) |
| D5   | No $32 - 7 = 25$ . So the difference is 25" ... "No. His total running time was 25 seconds. You need to subtract the amount of time it took him to run up the first half of the hill to find your answer — Telling moves are labeled and visible in transcript; student resolves confusion only after explicit revelation                                                                                                                                                                                                       |
| D23  | Teacher: (telling)You need to find out how many times 48 goes into 192" / "Teacher: (telling)192 divide by 48 =?" / "Teacher: (telling)4 is the correct answer you don't need to calculate further — Three consecutive Telling moves before student accepts correct answer; demonstrates reluctant resort to direct instruction                                                                                                                                                                                                 |

## Information Gaps

- Whether the 'providing guidance' dimension the user prioritized [A3] is operationally equivalent to the benchmark's Coherence+Correctness combination, or to Equitable Tutoring, or to none of these

## Requires Expert Verification

- Whether Indian metro teachers would re-classify dialogues like DATASET-D12 and DATASET-D28 (Generic openings that pre-solve) as Telling — testing operational stability of the taxonomy across populations
- Optimal reweighting of the three human-evaluation dimensions for Indian pedagogical priorities

## Remediation

**Gap 1: Telling@k frames direct instruction as a failure mode and Equitable Tutoring rewards withholding answers — both encode constructivist values misaligned with Indian professional teacher priorities. 'Providing guidance' (user-confirmed primary dimension) is not commensurately weighted.**

**Recommendation:** Replace Telling@k with a 'Guidance Adequacy@k' metric calibrated by Indian teacher judgments of when directive telling is appropriate. Reweight the three human-evaluation dimensions (Coherence, Correctness, Equitable Tutoring) to give 'Providing Guidance' (operationalized as adequacy and timeliness of directive instruction) primary weight per user elicitation.

# Output Content (OC)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: MATHDIAL | Context: Indian Metropolitan Professional Teachers (Grades 1–8)

**Output Content definition:** Whether ground-truth labels align with the judgments of regional stakeholders.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

## Justification

User-flagged HIGH risk and the highest-priority validity concern. Ground-truth quality judgments were produced by 91 professional teachers recruited via Prolific [\[Q23\]](#), predominantly UK nationals, followed by USA, Canada, Australia, India, Germany [\[Q27\]](#) — India is the fifth of six named countries with no disaggregated count reported. Annotators were explicitly instructed not to correct student solutions by telling but to give students opportunities to explore [\[Q148\]](#), systematically encoding Western constructivist norms regardless of annotator origin. Inter-annotator agreement is concerning:  $\kappa=0.60$  between independent annotators but only  $\kappa=0.49$  and  $\kappa=0.34$  between annotators and original teacher for move labeling [\[Q69\]](#), and critically  $\kappa=0.34$  for Equitable Tutoring [\[Q100\]](#) — the dimension most sensitive to cross-cultural variation. The user explicitly expects systematic divergence between Indian professional teachers' judgments and the benchmark's annotator-derived ground truth, particularly on direct-correction vs. Socratic probing and exam-readiness weighting [\[A4\]](#). Empirically, the dataset shows pervasive Socratic-mode dialogues with multiple Focus/Probing moves before any Telling [\[DATASET-D2, DATASET-D3\]](#), gentle non-directive corrections [\[DATASET-D20\]](#), and 'had to reveal' framing on successful Telling resolutions [\[DATASET-D5, DATASET-D23\]](#) — patterns the user identifies as misaligned with Indian professional norms. Web research corroborates: PARIKSHA found LLM evaluators align less with human evaluations on culturally nuanced responses [\[WEB-18\]](#), and systematic review confirms majority Indic datasets are translations with poor socio-cultural alignment [\[WEB-15\]](#). No India-specific math tutoring dialogue benchmark exists [\[WEB-15, WEB-16, WEB-17, WEB-18\]](#). This violates both convergent validity (labels do not correlate with regional perspectives) and external validity (judgments do not generalize).

| ID                   | Evidence                                                                                                                                                                                                                                                                                                                                                                       |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">Q23</a>  | "We recruit professionals with teaching experience through Prolific. We only select teachers who have completed at least 500 submissions and achieved a 100% completion rate." (p.4)                                                                                                                                                                                           |
| <a href="#">Q27</a>  | "The majority of annotators are nationals of the UK, followed by the USA, Canada, Australia, India, and Germany, with a median age of 39 years." (p.4)                                                                                                                                                                                                                         |
| <a href="#">Q69</a>  | "We obtain an agreement of $\kappa = 0.60$ between the two annotators and $\kappa = 0.49$ and $\kappa = 0.34$ , respectively, between either of the annotators and the teacher." (p.6)                                                                                                                                                                                         |
| <a href="#">Q100</a> | "We obtain agreements of $\kappa = 0.29$ , $\kappa = 0.69$ , and $\kappa = 0.34$ for the three categories." (p.8)                                                                                                                                                                                                                                                              |
| <a href="#">Q148</a> | "Specifically, they were instructed to not just correct student solutions by telling what's correct/incorrect, but to give students the opportunity to explore the problem with a focus on core aspects, such as their chosen strategy." (p.17)                                                                                                                                |
| <a href="#">D2</a>   | Hi Rishi. Can you tell me how many hours it would take Apple to run 24 miles?" ... "Good. How many hours would it take Mac?" ... "How much longer does it take Apple?" ... "How much longer is that in minutes?" ... "So how much faster is Mac than Apple in minutes?" — Six consecutive probing questions guiding student to answer; entirely Socratic, no direct correction |
| <a href="#">D3</a>   | Carmela has $600 \times 2 + 50 = 1250$ " ... "It is 1250. Not 2450" ... " $600+950+1250 = 2800$ — Teacher resorts to Telling only after multiple Focus moves fail; Telling framed as reluctant fallback                                                                                                                                                                        |
| <a href="#">D5</a>   | No $32-7 = 25$ . So the difference is 25" ... "No. His total running time was 25 seconds. You need to subtract the amount of time it took him to run up the first half of the hill to find your answer — Telling moves are labeled and visible in transcript; student resolves confusion only after explicit revelation                                                        |
| <a href="#">D20</a>  | Teacher: I feel you have made this more complicated than needed could you maybe find a simpler way — Generic, mild correction without directing student to specific error; low-directive style                                                                                                                                                                                 |
| <a href="#">D23</a>  | Teacher: (telling)You need to find out how many times 48 goes into 192" / "Teacher: (telling)192 divide by 48 =" / "Teacher: (telling)4 is the correct answer you dont need to calculate further — Three consecutive Telling moves before student accepts correct answer; demonstrates reluctant resort to direct instruction                                                  |



|               |                                                                                                                                                                                                          |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>WEB-15</b> | Systematic review: majority Indic LLM datasets are translations and models fare poorly on socio-cultural tasks ( <a href="https://arxiv.org/html/2501.13912v1">https://arxiv.org/html/2501.13912v1</a> ) |
| <b>WEB-16</b> | IndicEval covers Indian high-stakes exams but is not a pedagogical dialogue benchmark ( <a href="https://arxiv.org/html/2602.16467">https://arxiv.org/html/2602.16467</a> )                              |
| <b>WEB-17</b> | Hindi LLM benchmarks suite focuses on math reasoning not tutoring quality ( <a href="https://arxiv.org/abs/2508.19831">https://arxiv.org/abs/2508.19831</a> )                                            |
| <b>WEB-18</b> | PARIKSHA finding: LLM evaluators align less with human evaluations on culturally nuanced responses ( <a href="https://arxiv.org/html/2406.15053v1">https://arxiv.org/html/2406.15053v1</a> )             |

## Strengths

- Annotators are professional teachers with vetted credentials (500+ submissions, 100% completion, training tests) [Q23, Q24, Q25, Q132]
- India represented at all in the annotator pool [Q27]
- Authors transparently report inter-annotator agreement statistics including low agreement on Equitable Tutoring [Q100]
- Self-correctness annotations enable filtering by resolution strategy [DATASET-D5]

| ID          | Evidence                                                                                                                                                                                                                                                                                                             |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Q23</b>  | "We recruit professionals with teaching experience through Prolific. We only select teachers who have completed at least 500 submissions and achieved a 100% completion rate." (p.4)                                                                                                                                 |
| <b>Q24</b>  | "Annotators read guidelines for the task in an initial training phase (cf. Section D.3) and then complete a test on an example conversation to assess their understanding of the task." (p.4)                                                                                                                        |
| <b>Q25</b>  | "We only select annotators with 100% test scores for further rounds of data collection, similar to Zhang et al. (2023)." (p.4)                                                                                                                                                                                       |
| <b>Q27</b>  | "The majority of annotators are nationals of the UK, followed by the USA, Canada, Australia, India, and Germany, with a median age of 39 years." (p.4)                                                                                                                                                               |
| <b>Q100</b> | "We obtain agreements of $\kappa = 0.29$ , $\kappa = 0.69$ , and $\kappa = 0.34$ for the three categories." (p.8)                                                                                                                                                                                                    |
| <b>Q132</b> | "To ensure the data quality we filter only annotators with 100% completion rate with more than 500 total submissions." (p.16)                                                                                                                                                                                        |
| <b>D5</b>   | No 32-7 = 25. So the difference is 25" ... "No. His total running time was 25 seconds. You need to subtract the amount of time it took him to run up the first half of the hill to find your answer — Telling moves are labeled and visible in transcript; student resolves confusion only after explicit revelation |

## Checklist

### Checklist item definitions:

| ID          | Assessment Question                                               |
|-------------|-------------------------------------------------------------------|
| <b>OC-1</b> | Do ground-truth labels reflect regional stakeholder perspectives? |
| <b>OC-2</b> | Assess potential annotator-regional population disagreement       |
| <b>OC-3</b> | Review annotator demographics from documentation                  |
| <b>OC-4</b> | Consider label re-annotation by regional annotators               |
| <b>OC-5</b> | Review aggregation methods for minority perspective erasure       |
| <b>OC-6</b> | Document label issues harming convergent/external validity        |

### Checklist responses:

**OC-1:** No — ground-truth labels reflect predominantly UK/US pedagogical perspectives [Q27] with annotators explicitly instructed toward Socratic exploration [Q148], not Indian metro professional teacher perspectives.

**OC-2:** Yes — substantial disagreement is expected. The user explicitly anticipates systematic divergence on direct-correction vs. Socratic probing and exam-readiness weighting [A4]. PARIKSHA documents LLM-human alignment gaps on culturally nuanced responses [WEB-18].

**OC-3:** Annotator demographics partially documented: 91 teachers, 71 female/18 male, median age 39, predominantly UK nationals [Q26, Q27]. Indian representation is not disaggregated [annotator\_provenance gap].

**OC-4:** Re-annotation by Indian metro professional teachers is strongly warranted given user-anticipated systematic divergence [A4].

**OC-5:** Aggregation methods ( $\kappa$  for inter-annotator agreement) [Q69, Q100] do not separately surface minority-country perspectives. With Indian annotators a small minority, their judgments would be erased in any majority-aggregation procedure.

**OC-6:** Convergent validity violation: Western-norm labels unlikely to correlate with Indian teacher judgments [A4, WEB-18].  
External validity violation: original judgments will not generalize to Indian metro context [WEB-15].

| ID     | Evidence                                                                                                                                                                                                                                        |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q26    | "We employ 91 expert annotators, of which 71 identify as female and 18 as male." (p.4)                                                                                                                                                          |
| Q27    | "The majority of annotators are nationals of the UK, followed by the USA, Canada, Australia, India, and Germany, with a median age of 39 years." (p.4)                                                                                          |
| Q69    | "We obtain an agreement of $\kappa = 0.60$ between the two annotators and $\kappa = 0.49$ and $\kappa = 0.34$ , respectively, between either of the annotators and the teacher." (p.6)                                                          |
| Q100   | "We obtain agreements of $\kappa = 0.29$ , $\kappa = 0.69$ , and $\kappa = 0.34$ for the three categories." (p.8)                                                                                                                               |
| Q148   | "Specifically, they were instructed to not just correct student solutions by telling what's correct/incorrect, but to give students the opportunity to explore the problem with a focus on core aspects, such as their chosen strategy." (p.17) |
| WEB-15 | Systematic review: majority Indic LLM datasets are translations and models fare poorly on socio-cultural tasks ( <a href="https://arxiv.org/html/2501.13912v1">https://arxiv.org/html/2501.13912v1</a> )                                        |
| WEB-18 | PARIKSHA finding: LLM evaluators align less with human evaluations on culturally nuanced responses ( <a href="https://arxiv.org/html/2406.15053v1">https://arxiv.org/html/2406.15053v1</a> )                                                    |

## Information Gaps

- Disaggregated count of Indian annotators in the pool [annotator\_provenance unresolvable from public sources]
- Whether any sampled dialogues were authored by Indian-origin annotators (dataset does not expose annotator identity per dialogue)
- Magnitude of expected  $\kappa$  between Indian metro teachers and benchmark ground truth on Equitable Tutoring

## Requires Expert Verification

- Whether re-annotation of a representative subsample by Indian metro teachers produces label distributions that diverge significantly from MATHDIAL ground truth on Equitable Tutoring
- Whether Indian teachers would re-rate currently-low-rated directive dialogues as high-quality

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## Remediation

**Gap 1: Ground-truth quality labels are dominated by UK/US annotators explicitly instructed toward Socratic non-direction, with  $\kappa=0.34$  on Equitable Tutoring even within the Western pool. Systematic divergence from Indian professional teachers' judgments expected.**

**Recommendation:** Re-annotate a stratified sample ( $\geq 100$  dialogues, balanced across self-correctness outcomes) with at least 5–10 Indian metro professional teachers per item. Compute  $\kappa$  between Indian and original annotators per dimension; report dimension-level disagreement and use the Indian-annotated subset as the operative ground truth for deployment evaluation.

**Gap 2: Dataset does not expose annotator identity per dialogue, so it is impossible to filter to dialogues authored by Indian-origin annotators or audit demographic representativeness at the item level.**

**Recommendation:** Request annotator-disaggregated metadata from the benchmark authors, or treat the entire dataset as Western-norm-anchored and weight evaluation results accordingly. Flag the unknown Indian-annotator contribution as a residual risk in deployment documentation.

# Output Form (OF)

Score: 5/5 — Strong alignment | Confidence: high

Benchmark: MATHDIAL | Context: Indian Metropolitan Professional Teachers (Grades 1–8)

**Output Form definition:** Whether the expected output modality matches regional deployment needs and accessibility.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

## Justification

Output form is well-aligned. Both benchmark and deployment evaluate text-based open-ended tutoring dialogue outputs. Automatic metrics (sBLEU, BERTScore, KF1, Uptake) [Q89, Q90, Q93] and interactive metrics (Success@k, Telling@k) [Q95, Q96] operate on text. Human evaluation uses 3-point Likert and binary scales [Q167] in standard formats. Authors transparently acknowledge automatic faithfulness metrics are insufficient [Q91, Q92] and that immediate solving success does not capture long-term learning [Q114]. The deployment context (English text-only on mobile/enterprise platforms) matches output modality completely. No literacy or accessibility issues for the target population (English-fluent professional teachers per population characteristics).

| ID   | Evidence                                                                                                                                                                                                                                                                  |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q89  | "We assess our models using the sacrebleu (Post, 2018) implementation of BLEU (sBLEU) (Papineni et al., 2002), as well as BERTScore between generated response ( $u_T + 1$ ) and annotated response ( $u^*_T + 1$ ) for each teacher response in the conversation." (p.7) |
| Q90  | "Furthermore, in line with previous works (Dziri et al., 2022; Daheim et al., 2023), we report BERTScore and the token level F1 (KF1) between generated utterance and math word problem as a proxy for faithfulness." (p.7)                                               |
| Q91  | "However, we note that an increase in these metrics can be caused by an increase in overlap, which may also indicate more telling and can be undesirable." (p.7)                                                                                                          |
| Q92  | "However, finding good evaluation metrics for assessing the faithfulness of dialogue tutors remains an open problem." (p.7)                                                                                                                                               |
| Q93  | "Finally, we measure the Uptake of the generated response (Demszky et al., 2021)." (p.7)                                                                                                                                                                                  |
| Q95  | "Success@k measures the percentage of conversations where the student reaches the correct final answer at least once within the first k turns (equivalent of % solve rate in prior work)." (p.7)                                                                          |
| Q96  | "Telling@k measures the percentage of conversations where the teacher explicitly tells the final answer before the student has reached it on their own within the first k turns." (p.7)                                                                                   |
| Q114 | "Finally, measuring a student's immediate success in solving a problem does not capture all the aspects of student learning. From a learning perspective, focusing on and measuring long-term learning is desired." (p.10)                                                |
| Q167 | "We use a 3-point Likert scale ranging from 1 (poor) and 3 (very good) for coherence and equitable tutoring and a binary scale for correctness." (p.18)                                                                                                                   |

## Strengths

- Output modality matches deployment exactly: text-based dialogue [Q89, Q95]
- Multiple complementary metrics provide multi-faceted evaluation [Q89, Q90, Q93, Q95, Q96]
- Authors transparent about metric limitations [Q91, Q92, Q114]
- Standard Likert/binary scales accessible to professional teacher evaluators [Q167]

| ID  | Evidence                                                                                                                                                                                                                                                                  |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q89 | "We assess our models using the sacrebleu (Post, 2018) implementation of BLEU (sBLEU) (Papineni et al., 2002), as well as BERTScore between generated response ( $u_T + 1$ ) and annotated response ( $u^*_T + 1$ ) for each teacher response in the conversation." (p.7) |
| Q90 | "Furthermore, in line with previous works (Dziri et al., 2022; Daheim et al., 2023), we report BERTScore and the token level F1 (KF1) between generated utterance and math word problem as a proxy for faithfulness." (p.7)                                               |

|             |                                                                                                                                                                                                                            |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Q91</b>  | "However, we note that an increase in these metrics can be caused by an increase in overlap, which may also indicate more telling and can be undesirable." (p.7)                                                           |
| <b>Q92</b>  | "However, finding good evaluation metrics for assessing the faithfulness of dialogue tutors remains an open problem." (p.7)                                                                                                |
| <b>Q93</b>  | "Finally, we measure the Uptake of the generated response (Demszky et al., 2021)." (p.7)                                                                                                                                   |
| <b>Q95</b>  | "Success@k measures the percentage of conversations where the student reaches the correct final answer at least once within the first k turns (equivalent of % solve rate in prior work)." (p.7)                           |
| <b>Q96</b>  | "Telling@k measures the percentage of conversations where the teacher explicitly tells the final answer before the student has reached it on their own within the first k turns." (p.7)                                    |
| <b>Q114</b> | "Finally, measuring a student's immediate success in solving a problem does not capture all the aspects of student learning. From a learning perspective, focusing on and measuring long-term learning is desired." (p.10) |
| <b>Q167</b> | "We use a 3-point Likert scale ranging from 1 (poor) and 3 (very good) for coherence and equitable tutoring and a binary scale for correctness." (p.18)                                                                    |

## Checklist

### Checklist item definitions:

| ID          | Assessment Question                                    |
|-------------|--------------------------------------------------------|
| <b>OF-1</b> | Does output modality match regional deployment needs?  |
| <b>OF-2</b> | Assess text-to-speech availability for speech output   |
| <b>OF-3</b> | Consider literacy rates and accessibility requirements |
| <b>OF-4</b> | Document form mismatches harming external validity     |

### Checklist responses:

**OF-1:** Yes — text-based open-ended dialogue output matches Indian metro deployment requirements (English text on mobile/enterprise platforms).

**OF-2:** Not applicable — deployment is text-only, no speech output required.

**OF-3:** Target population (professional credentialed teachers, English-fluent) has high literacy [population\_characteristics]; no accessibility issues from output form.

**OF-4:** No form mismatches identified. Note that scoring-rubric content (Equitable Tutoring framing) is an OO concern, not OF — the form itself (Likert scale on text) is appropriate.

| ID          | Evidence                                                                                                                                                                                                                                                               |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Q89</b>  | "We assess our models using the sacrebleu (Post, 2018) implementation of BLEU (sBLEU) (Papineni et al., 2002), as well as BERTScore between generated response ( $uT + 1$ ) and annotated response ( $u^T + 1$ ) for each teacher response in the conversation." (p.7) |
| <b>Q91</b>  | "However, we note that an increase in these metrics can be caused by an increase in overlap, which may also indicate more telling and can be undesirable." (p.7)                                                                                                       |
| <b>Q114</b> | "Finally, measuring a student's immediate success in solving a problem does not capture all the aspects of student learning. From a learning perspective, focusing on and measuring long-term learning is desired." (p.10)                                             |
| <b>Q167</b> | "We use a 3-point Likert scale ranging from 1 (poor) and 3 (very good) for coherence and equitable tutoring and a binary scale for correctness." (p.18)                                                                                                                |